



ECE3700J Introduction to Computer Organization

Lab 3 – Pipelined Processor

Purpose

This lab is intended to help you better understand the concepts of pipelined architecture as well as how different instructions are executed by a pipelined processor. For simplicity purpose, we assume the hazard problems are all resolved by stalls in software, so your modeled hardware doesn't have to worry about hazards.

Tasks

This lab is similar to and based on the single-cycle processor lab, except you will model a pipelined processor in Verilog HDL that executes a bigger subset of RISC-V instructions including:

- The arithmetic-logical instructions `add`, `addi`, `sub`, `and`, `andi`, `or`, `sll`, `slli`, `srl`, `srlr`, and `sra`
- The memory-reference instructions `lw`, `sw`, `lb`, `lbu`, and `sb`
- The jumping instructions `beq`, `bne`, `bge`, `blt`, `jal`, and `jalr`

Your modules must be simulated with a Verilog simulator and synthesized by using Xilinx synthesis tool. This means your Verilog code must be **synthesizable**. A simple RISC-V assembly testing program without any hazard will be provided to verify that your processor can execute those instructions continuously and correctly.

FPGA hardware implementation is NOT required for this lab.

Team Organization

This lab is a team effort. Each team should consist of 3 students, randomly grouped. The work should be appropriately divided and distributed among all team members. Every student will be given oral examination upon demonstration. Students are not allowed to switch teams without permission of the instructor.

Deliverables

- **Demonstration** – Every team should demonstrate to the teaching group the following before your lab session ends:
 - 1) Simulation results of the top-module of your design
 - 2) RTL schematic of your Verilog design generated with Xilinx Vivado software.Each team member should be prepared for an oral exam on this lab during the demonstration.
- **Lab report** – The lab report should be a written report including:
 - 1) Brief description of all aspects of the modeling and implementation of the processor;
 - 2) Screen shots and brief explanations of simulation results for each of the instructions: `add`, `addi`, `lw`, `sw`, `beq`, and `jal`.
 - 3) RTL schematic of your Verilog model generated with Xilinx Vivado software;



- **Peer Evaluation** – Each team member is required to provide a peer evaluation for the team effort in this lab. The marks of the peer evaluation should be integers ranging between 0 to 10, inclusively, with 10 indicating the biggest contribution. A mark should be given to each team member including yourself according to the team member's contribution based on your observation. A brief description of contribution of each team member should also be provided, as shown in the following table.

Name	Level of contribution (0 ~ 10)	Description of contribution
(yourself)		
(your lab partner 1)		
(your lab partner 2)		

- **Source Files** – All your Verilog source files and any other supporting files.

This is a 3-week lab. The full score for this lab is 500 points. All required documents should be submitted on Canvas before **22:00pm, July 5, 2025**.

Grading

- Lab report: 30%
- Demonstration: 50%
 - Working Verilog model (simulation): 30%
 - Individual oral exam: 20%
- Source files and peer evaluation: 20%